

A Byzantine-Robust Federated Data-Driven Evolutionary Dynamic Optimization Framework for Secure Financial Edge Networks

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MSc Artificial Intelligence

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June 2nd, 2026

1 Introduction

Financial services are rapidly migrating computation to the network edge—in fraud detection, high-frequency portfolio rebalancing, and real-time risk management—where decisions must be optimised under continuously shifting market conditions. These are inherently *dynamic optimisation problems* (DOPs): the fitness landscape changes over time and the algorithm must *track a moving optimum* rather than converge to a static solution [1, 2]. Two unsolved problems obstruct deployment. First, evaluating candidate solutions on live financial infrastructure is prohibitively expensive, motivating *data-driven* (surrogate-assisted) evolutionary optimisation [3]. Second, edge nodes are geographically distributed and untrusted, requiring a *federated* design to avoid centralising regulated data—yet federated systems are acutely vulnerable to *Byzantine* participants (malicious or faulty nodes) that can poison the shared surrogate model without detection [4, 5]. No existing framework unifies these three requirements. This three-year PhD project designs, implements, and evaluates a Byzantine-robust federated data-driven evolutionary dynamic optimisation (BF-DDEDO) framework, develops a novel drift-aware robust aggregation operator, and produces deployment guidance for autonomous optimisation in regulated financial infrastructure. A mixed-methods design combining algorithm construction, benchmarking, and practitioner validation is used throughout.

2 Background

Evolutionary algorithms extended with diversity-maintenance, memory, and prediction mechanisms have been studied extensively for DOPs [1]. A persistent challenge is the *expensive evaluation* setting: assessing a single candidate portfolio or resource-allocation policy against live market data carries significant cost. Data-driven evolutionary optimisation [3] addresses this by building surrogate models—Gaussian processes, radial basis function networks, or neural networks—that approximate the costly objective. Combining surrogate assistance with non-stationary landscapes, where the surrogate itself must adapt as the objective drifts, remains an open problem with few systematic treatments.

Federated learning [6] enables collaborative model construction across distributed nodes without transmitting raw data, satisfying data-residency requirements of financial regulation. However, federated aggregation is vulnerable to Byzantine participants: a minority of malicious nodes can corrupt a naively averaged global model arbitrarily. Robust aggregation rules—Krum [4] and coordinate-wise trimmed-mean/median [5]—provide guarantees against a bounded Byzantine fraction only in the *static, gradient-based* setting. Their interaction with *population-based evolutionary search under concept drift* is uncharted: when the landscape shifts legitimately, node updates change abruptly—a signal indistinguishable from adversarial poisoning under standard robust aggregators [8]. Kairouz et al. [7] catalogue the federated threat surface but do not address evolutionary or surrogate-based settings. The Bank of England and FCA [9] have explicitly called for robust, explainable AI in financial decision-making, creating direct regulatory demand. This confluence of open technical problem, regulatory imperative, and deployment pressure defines a precise, timely, and publishable research gap.

3 Proposed Work

3.1 Aims

Aim: To design, implement, and rigorously evaluate a Byzantine-robust federated framework for data-driven evolutionary dynamic optimisation that enables secure and adaptive optimisation in distributed financial edge environments subject to concept drift and adversarial participants, and to validate its practical deployment readiness with domain experts in financial AI and cybersecurity.

3.2 Objectives

1. Conduct a systematic literature review across evolutionary dynamic optimisation, surrogate-assisted optimisation, federated learning, and Byzantine-robust aggregation to formally characterise the research gap.
2. Construct a benchmark problem suite combining the Moving Peaks Benchmark [10] and GDBG [11] with a parameterised financial-edge simulation, varying drift severity, frequency, and Byzantine fraction.
3. Design and implement a federated data-driven EDO architecture in which edge nodes maintain local surrogate models and collaborate via a secure aggregation protocol.
4. Develop a novel drift-aware Byzantine-robust aggregation operator that statistically distinguishes legitimate environmental drift from adversarial model poisoning, and establish tolerance for a bounded malicious fraction f .
5. Empirically evaluate BF-DDEDO against non-robust and existing-robust baselines (Krum, coordinate-wise median/trimmed-mean), measuring offline tracking error, convergence speed, communication overhead, and robustness degradation.
6. Validate the framework's practical deployment readiness through questionnaire and interview studies with financial AI practitioners and produce deployment guidance addressing legal and regulatory requirements.

3.3 Sub-Objectives

1. Identify and compare surrogate model architectures (Gaussian processes, RBF networks, neural networks) suited to non-stationary federated optimisation.
2. Determine Byzantine attack strategies most relevant to population-based evolutionary search, distinguishing gradient-based threats from surrogate-poisoning strategies.
3. Establish communication-efficiency and latency requirements for BF-DDEDO deployment, identifying acceptable overhead thresholds for high-frequency financial decisions.
4. Assess practitioner awareness of Byzantine threats in financial edge AI and identify real-world compliance gaps via questionnaire and interview data.
5. Map provisions of UK GDPR, Data Protection Act 2018, FCA regulations, and EU AI Act 2024 that apply to autonomous optimisation in financial edge infrastructure to framework design requirements.

4 Rationale

Three converging trends make this research timely and necessary. First, the FCA–BoE joint discussion paper [9] identifies Byzantine robustness and explainability as open requirements for AI in financial systems, without an existing solution for the evolutionary, surrogate-assisted, non-stationary setting. Second, edge AI in finance is accelerating from pilot to production, exposing the federated attack surface to compromised trading terminals and IoT risk-sensor nodes that move real capital. Third, the methodological gap is

precise: while Byzantine-robust federated learning and evolutionary dynamic optimisation are each mature subfields, their intersection—where legitimate drift confounds adversarial detection—is essentially absent from the literature. The proposed drift-aware aggregation operator constitutes a genuine methodological contribution of independent interest to both the evolutionary computation and trustworthy-AI communities, well beyond the financial application domain. The practitioner validation component additionally addresses the acknowledged disconnect between academic AI robustness research and real-world financial deployment practice.

5 Methodology

The project follows a *constructive, experimental, mixed-methods* research paradigm. The primary strand designs, implements, and validates algorithms and framework components through controlled computational experiments with statistical rigour, consistent with established practice in evolutionary computation [1, 3]. A secondary qualitative–quantitative strand collects practitioner evidence through questionnaires and semi-structured interviews to validate deployment readiness and identify real-world requirements not captured by simulation. The two strands are integrated in WP5–WP6. The computational strand proceeds through four stages: (1) *Formalisation* of the federated data-driven DOP, drift model, surrogate formulation, and adversary model; (2) *Algorithm design*, producing the novel drift-aware robust aggregator that compares each node’s update against a global drift-trajectory reference; (3) *Benchmarking* on Moving Peaks and GDBG before a financial-edge simulation provides ecological validity; and (4) *Statistical validation* using ≥ 30 independent runs with Wilcoxon rank-sum tests, Friedman post-hoc analysis, and effect sizes.

5.1 Questionnaires

An online questionnaire will be distributed to financial AI practitioners and cybersecurity professionals ($n \geq 30$). It covers: (i) participant background; (ii) awareness of Byzantine threats in federated AI; (iii) ranking of deployment requirements for robust federated optimisation; and (iv) perceived usability of the BF-DDED framework from a provided summary. Responses yield Likert-scale quantitative data and open-text qualitative data analysed thematically. Distribution is via LinkedIn and fintech forums over 6 weeks in Year 3. DMU Research Ethics Committee approval will be obtained; participation is voluntary and anonymised.

5.2 Interviews

Semi-structured interviews ($n = 10\text{--}15$) are drawn from questionnaire respondents willing to participate further, selected to represent diverse roles (developer, compliance officer, risk manager, regulator). Interviews explore practitioner experience, regulatory concerns, and technical requirements for financial edge deployment in depth. Each lasts approximately 45 minutes via video conference, is recorded with consent, transcribed verbatim, and analysed using Braun and Clarke’s reflexive thematic analysis. Findings are synthesised with computational results to produce the final deployment guidance framework.

6 Work Packages

WP1 – Literature Review and Gap Formalisation (Months 1–6). Systematic review of EDO, surrogate-assisted optimisation, federated learning, and Byzantine robustness. Formally characterises the gap; fixes evaluation metrics. *Linked to O1, SO1–SO2. Deliverables:* survey manuscript; gap report.

WP2 – Problem Formulation and Benchmark Suite (Months 5–12). Formal DOP, drift model and adversary model definition. Moving Peaks/GDBG benchmark and financial-edge simulation construction. *Linked to O2, SO3. Deliverables:* open benchmark suite; technical report.

WP3 – Federated Data-Driven EDO Architecture (Months 11–19). Distributed architecture with local surrogate models, local EA, federated protocol, and non-robust aggregation baseline. *Linked to O3, SO1. Deliverables:* working prototype; conference paper.

WP4 – Byzantine-Robust Aggregation Operator (Months 18–27). Drift-aware robust aggregator design; theoretical/empirical tolerance analysis; framework integration. *Linked to O4, SO2. Deliverables:* novel operator; journal paper.

WP5 – Experimental Evaluation and Practitioner Validation (Months 25–33). Comprehensive computational experiments vs. baselines; practitioner questionnaire ($n \geq 30$) and interviews ($n = 10\text{--}15$); thematic and statistical analysis. *Linked to O5–O6, SO3–SO4. Deliverables:* experimental dataset; evaluation paper; validated deployment guidance framework.

WP6 – Ethics/Legal Analysis, Thesis and Dissemination (Months 31–36). Legal/ethical compliance mapping; thesis writing and submission; open-access release of code, benchmarks, and results. *Linked to O6, SO5. Deliverables:* PhD thesis; legal compliance note; ≥ 2 journal papers.

7 Professional, Legal and Ethical Issues

Financial regulation: Optimisation influencing capital allocation falls within FCA and PRA regulatory perimeters. The FCA Senior Managers Regime requires human accountability for algorithmic decisions; the BF-DDED framework logs decision provenance and supports mandatory human oversight as an architectural requirement. No live market trading will be conducted during the research programme.

Data protection: The federated design avoids centralising regulated data. Simulation datasets are anonymised or fully synthetic. All research data are governed by UK GDPR and the Data Protection Act 2018, stored on DMU’s secure institutional repository with automated backup. Questionnaire and interview data are anonymised before analysis; participants may withdraw at any time.

Security and dual-use: Byzantine attack implementations are confined to the simulation environment and not released in weaponisable form. Responsible-disclosure principles apply to any vulnerabilities discovered in third-party federated libraries. All research intent and outputs are defensive.

Professional standards: The research follows ACM and BCS codes of conduct and DMU research-integrity policy. Open code, fixed random seeds, and documented benchmarks are treated as ethical obligations ensuring reproducibility.

8 Relevance to Beneficiaries

Financial institutions gain a method to optimise under changing conditions while maintaining robustness against compromised nodes, directly reducing operational and systemic risk. **Regulators** (FCA, BoE) receive evidence-based technical foundations for AI robustness guidance. **Scientific communities** benefit: the evolutionary computation community gains a new federated data-driven DOP problem class; the trustworthy-AI community gains a drift-aware Byzantine-robust technique with analysed guarantees.

Practitioners gain a validated deployment framework grounded in real-world requirements captured via questionnaire and interview. **Society** benefits from more resilient financial infrastructure. **Commercial and scientific risks:** the novel aggregator may not outperform existing robust methods under genuine drift—this is a publishable negative result, and the staged WP design ensures a baseline is established before the core contribution is attempted. Scope creep is mitigated by tight WP-to-objective linkage throughout.

9 Research Management Plan

The candidate will be supervised by a primary supervisor in evolutionary dynamic optimisation and a co-supervisor in distributed/secure systems. Fortnightly supervision meetings review progress and set next-period objectives. Formal milestone reviews at months 6, 18, and 30 comply with DMU Research Degree Regulations including the mandatory Year 1 progression review. All code and configurations are version-controlled via Git; manuscripts and data are stored on DMU’s institutional secure repository. *Risk register:* (1) *Computational cost* of large parallel experiments, mitigated by DMU HPC cluster access and staged experiment scaling; (2) *Theoretical intractability* of the tolerance proof, mitigated by an empirical-guarantee

fallback with comprehensive stress-testing; (3) *Low questionnaire/interview response*, mitigated by early engagement with fintech LinkedIn communities and professional networks; (4) *Third-party library restrictions*, mitigated by maintaining open-source simulation fallbacks throughout all work packages.

10 Justification of Resources

Personnel: One full-time PhD candidate with machine learning, optimisation, and software-engineering background. No additional personnel required.

Software: Python ecosystem (NumPy, scikit-learn, PyTorch, DEAP), Flower federated-learning framework, NVivo for qualitative interview analysis—all open-source or university-licensed. LaTeX for thesis. No commercial licences required.

Hardware: Standard development workstation plus DMU HPC/GPU cluster for the ≥ 30 independent runs per configuration across the full Byzantine fraction and drift-severity grid. No dedicated hardware purchase required.

Data: Publicly available financial time-series (Yahoo Finance, Kaggle) and synthetic edge workloads. Questionnaire/interview data collected with ethical approval; no personal financial data acquired.

Dissemination: Open-access APCs estimated at £1,500–£3,000; conference registration and travel approximately £1,500 over three years.

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Gantt Chart

Figure 1: Project Work Plan

Task	Description	Deliverable	Months
WP1	Literature Review & Gap Formalisation: EDO, surrogate optimisation, federated learning, Byzantine robustness	Survey paper; Gap report	6
WP2	Problem Formulation & Benchmark Suite: DOP model, adversary model, Moving Peaks, GDBG, financial-edge simulation	Open benchmark suite; Technical report	8
WP3	Federated Data-Driven EDO Architecture: local surrogate models, EA, federated protocol, non-robust baseline	Working prototype; Conference paper	9
WP4	Byzantine-Robust Aggregation Operator: drift-aware design, tolerance analysis, integration	Novel operator; Journal paper	10
WP5	Experimental Evaluation & Practitioner Validation: benchmarks, questionnaire ($n \geq 30$), interviews ($n = 10-15$)	Dataset; Evaluation paper; Guidance framework	9
WP6	Ethics/Legal Analysis, Thesis & Dissemination: compliance mapping, thesis writing, open release	PhD thesis; Legal note; ≥ 2 papers	6

Figure 2: Project Work Plan Diagram

Tasks	Year 1				Year 2				Year 3			
	1-3 Months	3-6 Months	6-9 Months	9-12 Months	1-3 Months	3-6 Months	6-9 Months	9-12 Months	1-3 Months	3-6 Months	6-9 Months	9-12 Months
Work Package 1												
Review EDO, surrogate & dynamic opt. literature												
Survey federated learning & Byzantine robustness												
Identify research gap; draft survey manuscript												
Work Package 2												
Define formal DOP, drift model & adversary model												
Construct Moving Peaks / GDBG benchmark suite												
Build financial-edge simulation & validate												
Work Package 3												
Design federated architecture & communication protocol												
Implement local surrogate model and evolutionary search												
Integrate modules & test non-robust baseline												
Work Package 4												
Design drift-aware robust aggregation operator												
Theoretical / empirical Byzantine tolerance analysis												
Integrate operator into framework & refine												
Work Package 5												
Controlled benchmark experiments (≥ 30 runs/config.)												
Financial-edge scenario evaluation vs. baselines												
Design & distribute practitioner questionnaire ($n \geq 30$)												
Conduct semi-structured interviews ($n=10-15$)												
Statistical & thematic analysis; guidance framework												
Work Package 6												
Final thesis submission & open dissemination												